

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA0509

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity. (BL2, BL3)*

Activity Tool : MCQ Test (Low)
Assessment Tool Weightage: 30%

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Which of the following is true about 1NF? (a) Allows multi-valued attributes (b) Eliminates partial dependencies (c) All attributes must be atomic (d) Eliminates transitive dependencies	BL2 – Understand	5
2	A relation is in 2NF if it is in 1NF and: (a) Has no transitive dependencies (b) Has no partial dependencies on primary key (c) Every attribute is a prime attribute (d) Has no multi-valued dependencies	BL2 – Understand	5
3	Which normal form deals with multi-valued dependencies? (a) 2NF (b) 3NF (c) BCNF (d) 4NF	BL2 – Understand	5
4	BCNF is a stricter version of: (a) 1NF (b) 2NF (c) 3NF (d) 4NF	BL2 – Understand	5
5	Functional dependency $X \rightarrow Y$ means: (a) Y determines X (b) X determines Y (c) X and Y are unrelated (d) Both determine each other	BL2 – Understand	5
6	Which of the following anomaly is NOT associated with unnormalized relations? (a) Insertion anomaly (b) Deletion anomaly (c) Update anomaly (d) Retrieval anomaly	BL2 – Understand	5
7	A superkey is: (a) A minimal set of attributes that uniquely identifies tuples (b) Any set of attributes that uniquely identifies tuples (c) An	BL2 – Understand	5

	attribute that depends on a non-prime attribute (d) A foreign key reference		
8	If R(A,B,C) has FD: $A \rightarrow B$ and $B \rightarrow C$, then C is: (a) Partially dependent on A (b) Fully dependent on A (c) Transitively dependent on A (d) Independently determined	BL2 – Understand	5
9	5NF is also known as: (a) Domain-Key Normal Form (b) Boyce-Codd Normal Form (c) Project-Join Normal Form (d) Element Normal Form	BL2 – Understand	5
10	Lossless-join decomposition ensures: (a) No data is lost during decomposition (b) All FDs are preserved (c) The relation is in BCNF (d) No null values exist	BL2 – Understand	5

Code of Conduct:

I Lokesh S (192511037) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

MCQ					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

Q1. Which of the following is true about 1NF?

- A. Allows multi-valued attributes
- B. Eliminates partial dependencies
- C. All attributes must be atomic
- D. Eliminates transitive dependencies

Answer: C. All attributes must be atomic

Explanation: 1NF requires every attribute to contain a single atomic value. Multivalued or repeating attributes are not allowed.

Q2. A relation is in 2NF if it is in 1NF and:

- A. Has no transitive dependencies
- B. Has no partial dependencies on primary key
- C. Every attribute is a prime attribute
- D. Has no multi-valued dependencies

Answer: B. Has no partial dependencies on primary key

Explanation: 2NF requires the relation to be in 1NF and to have no partial dependency of a non-prime attribute on part of a composite key.

Q3. Which normal form deals with multi-valued dependencies?

- A. 2NF
- B. 3NF
- C. BCNF
- D. 4NF

Answer: D. 4NF

Explanation: Fourth Normal Form (4NF) specifically deals with multivalued dependencies.

Q4. BCNF is a stricter version of:

- A. 1NF
- B. 2NF
- C. 3NF
- D. 4NF

Answer: C. 3NF

Explanation: Boyce-Codd Normal Form (BCNF) is a stricter version of Third Normal Form (3NF).

Q5. Functional dependency $X \rightarrow Y$ means:

- A. Y determines X
- B. X determines Y
- C. X and Y are unrelated
- D. Both determine each other

Answer: B. X determines Y

Explanation: The functional dependency $X \rightarrow Y$ means that the value of X uniquely determines the value of Y.

Q6. Which of the following anomaly is NOT associated with unnormalized relations?

- A. Insertion anomaly
- B. Deletion anomaly
- C. Update anomaly
- D. Retrieval anomaly

Answer: D. Retrieval anomaly

Explanation: Insertion, deletion, and update anomalies are the standard anomalies associated with poorly designed or unnormalized relations.

Q7. A superkey is:

- A. A minimal set of attributes that uniquely identifies tuples
- B. Any set of attributes that uniquely identifies tuples
- C. An attribute that depends on a non-prime attribute
- D. A foreign key

Answer: B. Any set of attributes that uniquely identifies tuples

Explanation: A superkey is any set of attributes that uniquely identifies each tuple in a relation. A candidate key is a minimal superkey.

Q8. If $R(A,B,C)$ has FD: $A \rightarrow B$ and $B \rightarrow C$, then C is:

- A. Partially dependent on A
- B. Fully dependent on A
- C. Transitively dependent on A
- D. Independently determined

Answer: C. Transitively dependent on A

Explanation: Since $A \rightarrow B$ and $B \rightarrow C$, C depends on A through B. Therefore, C is transitively dependent on A.

Q9. 5NF is also known as:

- A. Domain-Key Normal Form
- B. Boyce-Codd Normal Form
- C. Project-Join Normal Form
- D. Element Normal Form

Answer: C. Project-Join Normal Form

Explanation: Fifth Normal Form (5NF) is also known as Project-Join Normal Form (PJ/NF) and deals with join dependencies.

Q10. Lossless-join decomposition ensures:

- A. No data is lost during decomposition
- B. All FDs are preserved
- C. The relation is in BCNF
- D. No null values exist

Answer: A. No data is lost during decomposition

Explanation: A lossless-join decomposition ensures that the original relation can be reconstructed without losing information.